



TACHYCARDIA WITH PULSES

Effective: January 1, 2026

Replaces: January 1, 2025

1. Patient Care Goals

- 1.1. Maintain adequate oxygenation, ventilation, and perfusion
- 1.2. Control ventricular rate
- 1.3. Restore regular sinus rhythm in unstable patient
- 1.4. Search for underlying cause
 - 1.4.1. Medications (caffeine, diet pills, thyroid, decongestants)
 - 1.4.2. Drugs (cocaine, amphetamines)
 - 1.4.3. History of dysrhythmia
 - 1.4.4. Congestive heart failure (CHF)

2. BLS Treatment

- 2.1. Routine Medical Care – Adult **(700-S04)**
 - 2.1.1. **Oxygen** – titrate as appropriate
- 2.2. Treat for signs and symptoms of shock as necessary **(700-A10)**

3. ALS Treatment

- 3.1. **Vascular Access (IV)**, TKO, antecubital (AC) location is preferred
- 3.2. Obtain **12-Lead ECG** if patient condition allows

4. Sinus Tachycardia

- 4.1. Consider **250 ml Normal Saline** and monitor the patient

5. Stable Supraventricular Tachycardia (SVT)

- 5.1. SVT can be identified by a regular rate greater than 150 bpm, a QRS duration less than 0.12 seconds, and absent P waves
- 5.2. Consider vagal maneuver
 - 5.2.1. If vagal maneuvers are unsuccessful consider **250 ml Normal Saline** while the provider prepares Adenosine
- 5.3. **Adenosine 6 mg Rapid IV** followed by a **20 ml Normal Saline rapid flush**
- 5.4. If rhythm does not convert in two (2) minutes administer:
 - 5.4.1. **Adenosine 12 mg Rapid IV** followed by a **20 ml Normal Saline rapid flush**
 - 5.4.2. May repeat once if rhythm does not convert after two (2) minutes

6. Atrial Fibrillation with Rapid Ventricular Response

- 6.1. If patient presents with heart rate greater than 150 bpm in atrial fibrillation AND patient's systolic blood pressure (SBP) is greater than (>) 100 mmHg administer:
- 6.2. **Magnesium Sulfate 2 gm IV in 100 ml Normal Saline**, at a rate of 60 gtt/min (1 gtt/sec) over 15 minutes if SBP > 100 mmHg
 - 6.2.1. If patient reports significant pain at site of magnesium administration, slow the drip rate until pain is controlled
- 6.3. Atrial fibrillation rarely requires cardioversion in the field, as it is difficult to ascertain the onset of this rhythm. The risk of stroke needs to be considered prior to cardioversion.

7. Unstable Supraventricular Tachycardia (SVT)



- 7.1. SVT can be identified by a regular rate greater than 150 bpm, a QRS duration less than 0.12 seconds, and absent P waves
- 7.2. Consider sedation, if SBP > 100 mm Hg, administer:
 - 7.2.1. **Midazolam 2.5 mg slow IV**, may repeat once in 2 minutes, max total dose of 5 mg
- 7.3. **Synchronized Cardioversion** at 100, 150, 200 joules (or manufacturer's recommended values);
 - 7.3.1. Starting with lowest recommended energy setting;
 - 7.3.2. Each subsequent counter shock increasing in energy

8. Torsades (Polymorphic Ventricular Tachycardia) with Pulse

- 8.1. Administer **Magnesium Sulfate 2 gm IV** in **100 ml Normal Saline** at a rate of 60 gtts/min (1 gtt/sec) over 15 minutes if SPB > 100 mmHg
 - 8.1.1. If patient reports significant pain at site of magnesium administration, slow the drip rate until pain is controlled
- 8.2. If SBP less than (<) 90 mmHg, **Defibrillate** at manufacturer's suggested values (example, 100, 150, 200 joules)
 - 8.2.1. Starting with lowest recommended energy setting;
 - 8.2.2. Each subsequent counter shock increasing in energy
- 8.3. **Amiodarone** is not indicated for Torsades

9. Stable Ventricular Tachycardia with Pulse

- 9.1. Administer **Amiodarone Drip 150 mg** in **100 ml Normal Saline**, over ten (10) minutes, (100 gtts/min delivered by Macro (10 gtts drip set)

10. Unstable Ventricular Tachycardia with Pulse

- 10.1. Consider sedation, if SBP > 100 mmHg, administer:
 - 10.1.1. **Midazolam 2.5 mg slow IV**, may repeat once in 2 minutes, max total of 5 mg
- 10.2. **Synchronized Cardioversion** at 100, 150, 200 joules (or manufacturer's recommended values);
 - 10.2.1. Starting with lowest recommended energy setting,
 - 10.2.2. Each subsequent counter shock increasing in energy

11. Special Considerations

- 11.1. Immediate cardioversion is seldom needed for heart rate less than 150 beats per minute
- 11.2. Pre-cardioversion sedation should be used in the awake patient whenever possible, use caution if the patient is hypotensive

12. Key Documentation Elements

- 12.1. Initial rhythm and all rhythm changes
- 12.2. Time, dose, and response to medications given
- 12.3. Obtain monitor strips after each intervention
- 12.4. Patient weight
- 12.5. History of event supporting treatment of underlying causes



13. Tachycardia with Pulses Treatment Flowchart

